```
> restart;
> F(r):=(3*R^2)/(14*Pi*r^2) - R/(4*Pi*r)+1/(14*Pi);
```

$$F(r) := \frac{3\,R^2}{14\,\pi\,r^2} - \frac{R}{4\,\pi\,r} + \frac{1}{14\,\pi} \tag{1}$$

```
> G := 6.674300000*10^(-11);Rp := 6371000;Rsun:=6.9634*10^8;Mp :=
  5.9722*10^24;MS := 0.19885*10^31;MMW := 640*10^9*MS;MLY:=
  9.4605284*10^21;R:=3.9228*10^22;Rs := 7.766461840*10^25;metru:=
  1.05702341*10^(-16);
```

$$G := 6.674300000 \times 10^{-11}$$

$$Rp := 6371000$$

$$Rsun := 6.963400000 \times 10^{8}$$

$$Mp := 5.972200000 \times 10^{24}$$

$$MS := 1.988500000 \times 10^{30}$$

$$MMW := 1.272640000 \times 10^{42}$$

$$MLY := 9.460528400 \times 10^{21}$$

$$R := 3.922800000 \times 10^{22}$$

$$Rs := 7.766461840 \times 10^{25}$$

$$metru := 1.057023410 \times 10^{-16} \tag{2}$$

```
> Rs := 7.766461840*10^25;metru:=1.05702341*10^(-16);Newt1:=-2*Pi*
  G*MMW*MLY^2;Newt:=-2*Pi*G*MMW*R^2;Newt/(4/3*Pi*R^3);
```

$$Rs := 7.766461840 \times 10^{25}$$

$$metru := 1.057023410 \times 10^{-16}$$

$$Newt1 := -4.776633816 \times 10^{76}$$

$$Newt := -8.212653398 \times 10^{77}$$

$$-3.247927940 \times 10^{9} \tag{3}$$

```
> Eond:=-(4*Pi*G*MMW/(3.471439144*10^44))*(2*R)^2*int((3.471439144*
  10^44/r^2 - 5.677056820*10^21/r + 1/(14*Pi))*r^3,r=Rsun..2*R);
```

$$Eond := -7.000700567 \times 10^{123} \tag{4}$$

```
> Eond*3/(4*2^3*Pi*2^3*R^3);Eond*3/(4*2^3*Pi*2^3*5.73^3*Rs^3);#The
  density of Energy/volume => Joule/m^3 onduscular ;
```

$$-4.325979122 \times 10^{53}$$

$$-2.963052970 \times 10^{41} \tag{5}$$

```
> VolUL:=4/3*Pi*Rs^3;Newt/VolUL
```

$$VolUL := 1.962267665 \times 10^{78}$$

$$-0.4185287025 \tag{6}$$

```
> subs(R=3.9228*10^22,F(r));
```

$$\frac{3\,R^2}{14\,\pi\,r^2} - \frac{R}{4\,\pi\,r} + \frac{1}{14\,\pi} \tag{7}$$

```
> subs(R=3.9228*10^22,%);
```

$$\frac{1.049628658 \times 10^{44}}{r^2} - \frac{3.121665052 \times 10^{21}}{r} + \frac{1}{14\,\pi} \tag{8}$$

```
> Fg(r) := -%*r^2*4*Pi/3;
```

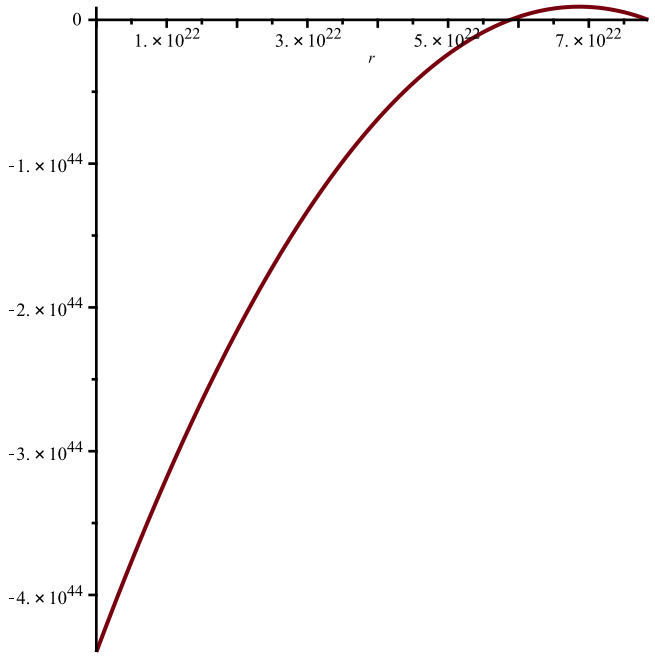
$$Fg(r) := -\frac{4\left(\frac{1.049628658 \times 10^{44}}{r^2} - \frac{3.121665052 \times 10^{21}}{r} + \frac{1}{14\,\pi}\right) r^2\,\pi}{3} \tag{9}$$

```
> R:=3.9228*10^22;int(Fg(r),r=Rp..2*R);
```

$$R := 3.922800000 \times 10^{22}$$

$$-9.581818760 \times 10^{66} \tag{10}$$

```
> plot(Fg(r),r = 1..2*R);
```




```
> h:= 6.626070150*10^(-34)/(2*Pi);c:= 299792458; rb:= 5.292410932*
  10^(-11);Sq := 4.30000000*10^(-19);Tq:= rb/c;PTQ:=Sq/c;
```

$$h := 1.054571818 \times 10^{-34}$$

$$c := 299792458$$

$$rb := 5.292410932 \times 10^{-11}$$

$$Sq := 4.300000000 \times 10^{-19}$$

$$Tq := 1.765358264 \times 10^{-19}$$

$$PTQ := 1.434325609 \times 10^{-27} \tag{11}$$

```
> Ev:=-h/(2*PTQ);V:=4/3*Pi*PTQ^3;DensitE:=Ev/V;DenEond:=Eond/(4/3*
  Pi*MLY^3);
```

$$Ev := -3.676193925 \times 10^{-8}$$

$$V := 1.236038126 \times 10^{-80}$$

$$DensitE := -2.974175187 \times 10^{72}$$

$$DenEond := -1.973816212 \times 10^{57} \tag{12}$$

```
> restart;
> Rs := 16.41866/2;_c2 := -9.711452983*c^2;#Billion Light Years
```

$$Rs := 8.209330000$$

$$_c2 := -9.711452983\, c^2 \tag{13}$$

```
> F(r):=(4*Rs^2*c^2 - 4*Rs^2*_c2)/(8*Pi*c^2)/r^2 - 1/4*Rs/(Pi*r) +
  _c2/(8*Pi*c^2);
```

$$F(r) := \frac{114.8904539}{r^2} - \frac{0.6532777245}{r} - 0.3864064366 \tag{14}$$

```
> F1(r):= evalf(int(F(r), r = 0.362 .. 2*13.79));
```

$$F1(r) := 299.8632335 \tag{15}$$

```
> F(r):=(-0.5939475112*ln(r) - 0.1201207224*r - 35.63280450/r)*4*
  Pi*r^2;
```

$$F(r) := 4\left(-0.5939475112\ln(r) - 0.1201207224\, r - \frac{35.63280450}{r}\right)\pi r^2 \tag{16}$$

```
> int(%,r=1..Rs);
```

$$-19017.97038 \tag{17}$$

```
> plot(F(r),r=1..2*Rs);
```

